

Unit 1.8 Inverse functions

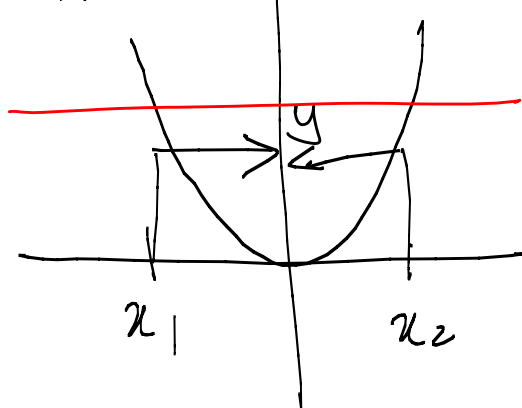
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Note Title

2015/02/11

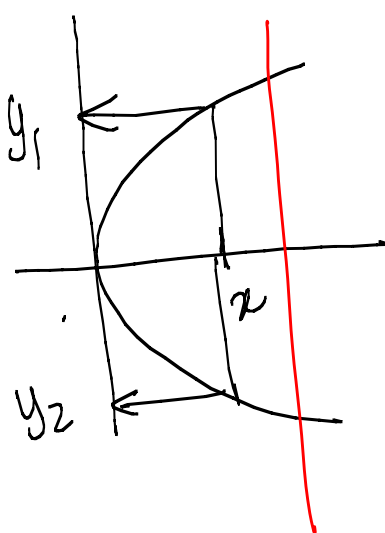
The concept of one-to-one.

Two to one

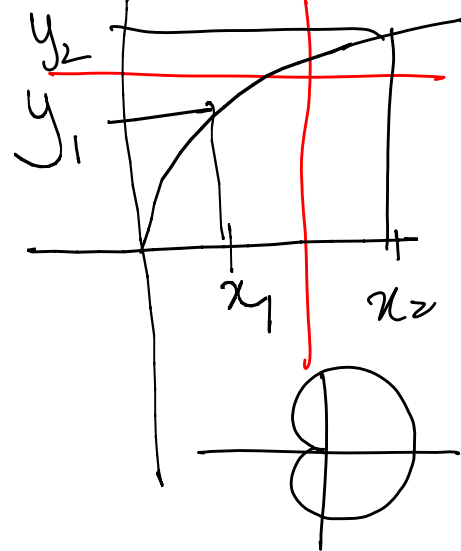


Function

One to two



one to one.



A one-to-one function passes both the vertical and horizontal line tests.

Definition:

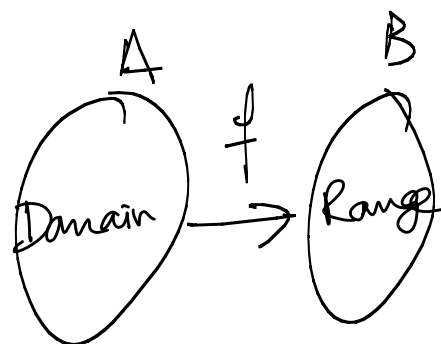
f is a one-to-one function.
if $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$

Another way of saying:

f is a one-to-one function
i.f $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

Notation:

$f: A \rightarrow B$
 $\{$ domain $\}$ range



NB
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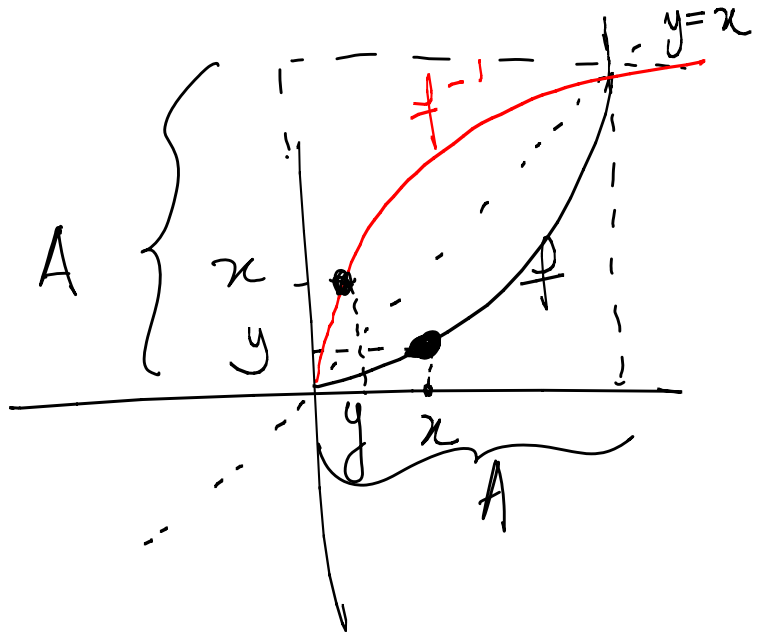
Definition of the inverse function.

Let f be a one-to-one function with domain A and range B

The inverse function of f , written f^{-1} is a function with domain B and range A such that

$$\underbrace{f^{-1}(y)} = x \iff \underbrace{y = f(x)}$$

Graphically:



Cancellation properties:

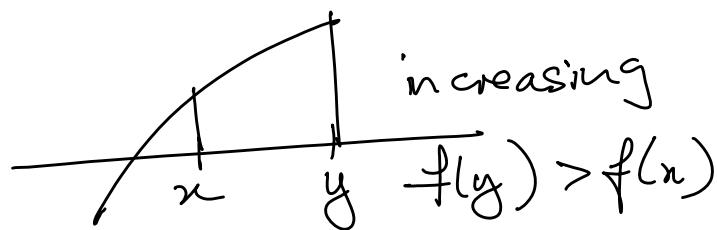
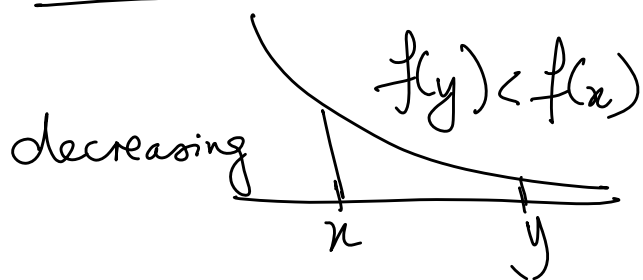
Let $f: A \rightarrow B$ and f is one-to-one

Then $f^{-1}: B \rightarrow A$ and

$$f \circ f^{-1}(x) = f(f^{-1}(x)) = x$$

$$f^{-1} \circ f(x) = f^{-1}(f(x)) = x.$$

Also:



Examples:

1. Inverse of $f(x) = \sqrt{x-1}$

$$y = \sqrt{x-1}$$

$$\Rightarrow x = \sqrt{y-1} \quad (\text{for inverse})$$

$$\Rightarrow x^2 = y-1$$

$$\Rightarrow y = x^2 + 1, \quad x \geq 0.$$

$$A = [1, \infty)$$

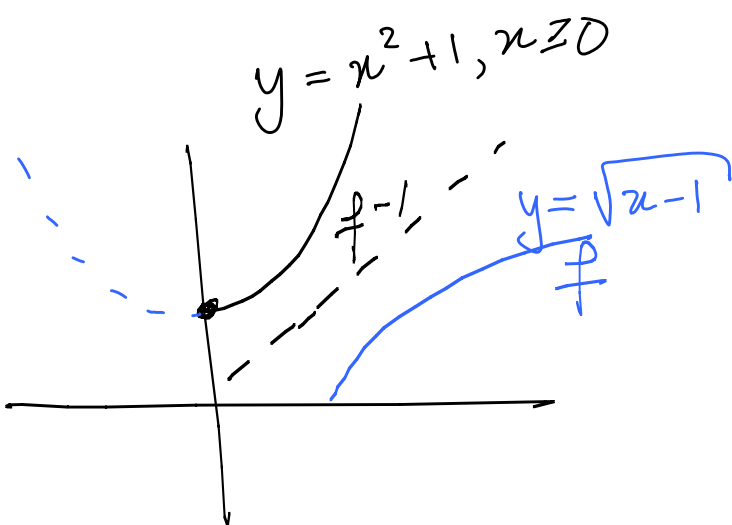
$$B = [0, \infty) \leftarrow$$

$$f : A \rightarrow B$$

$$\text{and } f^{-1} : B \rightarrow A$$

$$\text{Domain: } [0, \infty)$$

$$\text{Range: } [1, \infty)$$



$$f \circ f^{-1}(x) = f(f^{-1}(x)) \quad x \geq 0$$

$$= f(x^2 + 1)$$

$$= \sqrt{x^2 + 1 - 1}$$

$$= \sqrt{x^2}$$

$$= x, \quad x \geq 0.$$

$$\sqrt{x^2} = |x|$$

$$\begin{aligned}
 f^{-1} \circ f(x) &= f^{-1}(f(x)) \\
 &= f^{-1}(\sqrt{x-1}) \\
 &= (\sqrt{x-1})^2 + 1 \\
 &= x - 1 + 1 \\
 &= x, \quad x \geq 1.
 \end{aligned}$$

Eg. Find the inverse of $y = x^3 + 2$

For the inverse: $x = y^3 + 2$

$$\Rightarrow y = \sqrt[3]{x-2}$$

$$\begin{aligned}
 x &\in \mathbb{R} \\
 y &\in \mathbb{R}.
 \end{aligned}$$